Topic 7: API Integration Patterns & Authentication Methods

Institution: Tayana Academy | LMS Assessment & Hands-On Guide
Course: n8n Workflow Automation — Level 2 (Intermediate)
Topic / Assessment: Topic 7: API Integration Patterns & Authentication Methods
Workflow Name: `Topic 7 - API Integration Patterns & Authentication Methods`
Target Audience: Automation Engineers & Integration Developers
Difficulty Level: Intermediate
Platform Version: n8n version 1.0.0 or higher
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Repository: n8n-hands-on-level-2

1. Executive Summary & Objective

In enterprise application integration, external APIs enforce heterogeneous authentication and authorization protocols. While static services commonly rely on **API Key headers**, modern OAuth2 and identity providers (such as Okta, Azure AD, Auth0, and AWS Cognito) require dynamic **Bearer Tokens (JWT)** generated via Client Credentials grant exchanges. Production workflows must seamlessly navigate both paradigms without hardcoding ephemeral tokens or executing redundant authentication round-trips.

This assessment designs and implements a production-grade **API Integration Patterns & Authentication Methods** testbed in n8n. The workflow demonstrates:

1. **API Key Header Authentication Pattern:** Dispatches secure requests injecting `X-API-Key` headers paired with automated retry policies (`Retry on Fail: 3x` , 2000 ms backoff).
2. **Bearer Token Auto-Refresh Lifecycle Pattern:** Evaluates in-memory static token cache (`$getWorkflowStaticData('global')`), enforces a **60-second proactive expiration buffer**, requests fresh tokens via Client Credentials token exchange upon expiration, and caches tokens across consecutive workflow executions.
3. **Multi-Pattern Branch Routing & Telemetry:** Utilizes an n8n Switch node to isolate authentication patterns and converges into a standardized authentication audit logger.


1. Manual_Trigger ➔ Code_SetAuthMode
- Mode 0: API Key Header Pattern
- Mode 1: Bearer Token Lifecycle Pattern
2. Switch_AuthPattern (Rule 0 vs Rule 1)
Mode 0 (Output 0)
Mode 1 (Output 1)
3. HTTP_CallWithApiKey
Header: X-API-Key
Retry: 3x @ 2000ms
4. Code_CheckTokenExpiry
(Inspect static memory)
5. IF_TokenIsExpired?
Needs Refresh
Valid (Cached)
6. HTTP_RefreshAuth
(Client Creds POST)
7. Code_SaveFreshToken
(Update staticData)
8. HTTP_CallWithBearerToken
Authorization: Bearer <JWT>
9. Code_AuditAuthSuccess
(Standardized Receipt)

2. Canvas Architecture Overview

Workflow Canvas Architecture Diagram

```
flowchart TD
    A["🟢 Manual_Trigger<br/>(Trigger on demand)"] --> B["🎛️ Code_SetAuthMode<br/>(Emit testMode 0 & 1)"]
    B --> C{"🔀 Switch_AuthPattern<br/>(Rule 0 vs Rule 1)"}

    C -- "Output 0: API Key" --> D["🛡️ HTTP_CallWithApiKey<br/>(GET with X-API-Key Header)"]

    C -- "Output 1: Bearer Token" --> E["🧠 Code_CheckTokenExpiry<br/>(Check staticData & 60s buffer)"]
    E --> F{"❓ IF_TokenIsExpired<br/>(needsRefresh == true?)"}

    F -- "True (Expired/Missing)" --> G["🔄 HTTP_RefreshAuthToken<br/>(POST /auth/token Client Creds)"]
    G --> H["💾 Code_SaveFreshToken<br/>(Store Bearer JWT & expiresAt)"]
    H --> I["🔑 HTTP_CallWithBearerToken<br/>(GET with Bearer Header)"]

    F -- "False (Valid Token)" --> I

    D --> J["📋 Code_AuditAuthSuccess<br/>(Format Verification Receipt)"]
    I --> J

    subgraph S1["1. Test Suite Selector"]
        A
        B
        C
    end

    subgraph S2["2. Pattern A: Static API Key"]
        D
    end

    subgraph S3["3. Pattern B: Bearer Token Lifecycle"]
        E
        F
        G
        H
        I
    end

    subgraph S4["4. Diagnostics & Audit"]
        J
    end
```

Canvas Node Directory

Node Name	Node Type	Version	Operational Role & Authentication Protocol
`Manual_Trigger`	`manualTrigger`	`1.0`	Starts on-demand verification of the authentication test suite.
`Code_SetAuthMode`	`code`	`2.0`	Emits test cases for API Key (`testMode: 0`) and Bearer Token (`testMode: 1`).
`Switch_AuthPattern`	`switch`	`3.0`	Routes execution based on `testMode` into independent authentication tracks.
`HTTP_CallWithApiKey`	`httpRequest`	`4.2`	Dispatches GET request with header `X-API-Key: key_live_prod_2026` and 3x retry policy.
`Code_CheckTokenExpiry`	`code`	`2.0`	Evaluates `$getWorkflowStaticData('global')` timestamp against `Date.now() + 60000` .
`IF_TokenIsExpired`	`if`	`2.0`	Gating conditional routing to fresh token exchange (True) or cached token dispatch (False).
`HTTP_RefreshAuthToken`	`httpRequest`	`4.2`	Exchanges `client_id` and `grant_type: client_credentials` for fresh JWT access token.
`Code_SaveFreshToken`	`code`	`2.0`	Commits `access_token` and calculated `tokenExpiresAt` into persistent static memory.
`HTTP_CallWithBearerToken`	`httpRequest`	`4.2`	Dispatches GET request injecting dynamic header `Authorization: Bearer <token>` .
`Code_AuditAuthSuccess`	`code`	`2.0`	Converges authentication receipts, emitting unified diagnostic verification records.

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3. Prerequisites & Environment Setup

- **n8n Environment:** Version 1.0.0 or higher.
- **Zero External Developer Portal Setup:** All endpoints are tested using self-contained mock services:

Mock Authentication Server Options:

- **Option A: Free Cloud Mock API via DummyJSON & Postman Echo (Recommended)**
  - Token Generation: Endpoint returning `{ "access_token": "jwt_mock_access_token_sec99182", "token_type": "Bearer", "expires_in": 3600 }` .
  - Header Echo & Verification: Query `https://postman-echo.com/headers` to verify headers (`X-API-Key` , `Authorization: Bearer ...`).
- **Option B: Postman Mock Server**
  Deploy a Postman mock collection containing `POST /oauth/token` (200 OK) and `GET /protected/resource` (200 OK).
- **Option C: In-n8n Local Webhook Mock Server**
  Deploy a 2-endpoint mock workflow in your local n8n instance responding to `POST /mock/auth/token` and

`GET /mock/protected/resource`.

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4. Step-by-Step Practical Build Guide

Step 1: Main Trigger & Mode Selector (`Code_SetAuthMode`)

1. Create a workflow named `Topic 7 - API Integration Patterns & Authentication Methods`.
2. Place a `Manual_Trigger` node.
3. Connect a Code Node named `Code_SetAuthMode` running in **Run Once for All Items** mode:

```
// Set to 0 to test API Key, 1 to test Bearer Token with Auto-Refresh
return [
  { json: { testMode: 0, testName: "API Key Header Pattern" } },
  { json: { testMode: 1, testName: "Bearer Token Lifecycle Pattern" } }
];
```

Step 2: Multi-Branch Routing (`Switch_AuthPattern`)

Add an n8n Switch node:

- **Mode:** `Rules`
- **Rule 0 (Output 0):** `{{ $json.testMode }}` Equals `0` (Number) ➔ API Key Path
- **Rule 1 (Output 1):** `{{ $json.testMode }}` Equals `1` (Number) ➔ Bearer Token Path

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Step 3: Implement Branch 0 — API Key Header Authentication (`HTTP_CallWithApiKey`)

Connect an HTTP Request node to **Output 0**:

- **Method:** `GET`
- **URL:** `https://dummyjson.com/c/mock-protected-apikey`
- **Send Headers:** `ON`
  - Name: `X-API-Key`
  - Value: `key_live_prod_2026`
- **Settings Tab:**
  - `Retry on Fail` : `ON`
  - `Max Tries` : `3`
  - `Wait Between Tries` : `2000 ms`

Step 4: Implement Branch 1 — In-Memory Token Expiration Check (`Code_CheckTokenExpiry`)

Connect a Code Node to **Output 1** running in **Run Once for Each Item** mode:

```
const staticData = $getWorkflowStaticData('global');
const now = Date.now();
let token = staticData.bearerToken || null;
let expiresAt = staticData.tokenExpiresAt || 0;

// Needs refresh if missing or expiring within 60 seconds (proactive refresh)
const needsRefresh = !token || (expiresAt - now < 60000);

return {
  json: {
    hasToken: Boolean(token),
    needsRefresh: needsRefresh,
    cachedToken: token,
    expiresAt: expiresAt
  }
};
```

Step 5: Route Expiration Evaluation (`IF_TokenIsExpired`)

Connect an IF Node to `Code_CheckTokenExpiry` :

- **Condition:** Boolean `Equal`
- **Value 1:** `={{ $json.needsRefresh }}`
- **Value 2:** `{{ true }}`

Step 6: Request & Cache Fresh Token (`HTTP_RefreshAuthToken` & `Code_SaveFreshToken`)

1. **Refresh Request (`HTTP_RefreshAuthToken`):**
   Connect to the **True (Top)** branch of `IF_TokenIsExpired` :
   - **Method:** `POST`
   - **URL:** `https://dummyjson.com/c/mock-auth-token`
   - **Send Body:** `ON` (JSON)
   - **Body JSON:**

     ```
     {
       "client_id": "client_app_100",
       "grant_type": "client_credentials"
     }
     ```

2. **Store Fresh Token (`Code_SaveFreshToken`):**
   Connect a Code Node to `HTTP_RefreshAuthToken` :

```
const staticData = $getWorkflowStaticData('global');
const response = $input.item.json;

staticData.bearerToken = response.access_token || "bearer_mock_jwt_token_99182";
staticData.tokenExpiresAt = Date.now() + ((response.expires_in || 3600) * 1000);

return {
  json: {
    activeToken: staticData.bearerToken,
    expiresAt: staticData.tokenExpiresAt,
    refreshed: true
  }
};
```

Step 7: Converge & Dispatch with Bearer Token (`HTTP_CallWithBearerToken`)

Add an HTTP Request node named `HTTP_CallWithBearerToken` :

- **Connection A (Refreshed):** Connect output of `Code_SaveFreshToken` to `HTTP_CallWithBearerToken` .
- **Connection B (Cache Hit):** Connect the **False (Bottom)** branch of `IF_TokenIsExpired` directly to `HTTP_CallWithBearerToken` .
- **Parameters:**
  - **Method:** `GET`
  - **URL:** `https://dummyjson.com/c/mock-protected-bearer`
  - **Send Headers:** `ON`
    - Name: `Authorization`
    - Value: `={{ 'Bearer ' + ($json.activeToken || $json.cachedToken) }}`

Step 8: Build Authentication Audit Summary (`Code_AuditAuthSuccess`)

Connect both `HTTP_CallWithApiKey` and `HTTP_CallWithBearerToken` into `Code_AuditAuthSuccess` :

```
const resp = $input.item.json;
return {
  json: {
    authStatus: "AUTHENTICATION_VERIFIED",
    patternTested: resp.authMethod || ($node['Switch_AuthPattern']?.json?.testMode === 0 ? "API_KEY_HEADER
    serverConfirmed: resp.status || "AUTHORIZED",
    deliveryConfirmation: resp.message || "Access granted to protected endpoint",
    verifiedAt: new Date().toISOString()
  }
};
```

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5. Node-by-Node Settings Blueprint

Node ID	Node Name	Node Type	Critical Settings	Custom Expressions / Code
N7-01	`Manual_Trigger`	Manual Trigger	Trigger execution on demand	N/A
N7-02	`Code_SetAuthMode`	Code	Run Once for All Items	Emits test cases for API Key (0) and Bearer (1)
N7-03	`Switch_AuthPattern`	Switch	Mode: Rules, 0=API Key, 1=Bearer	Routes by `testMode`
N7-04A	`HTTP_CallWithApiKey`	HTTP Request	GET `/mock/protected/resource`, Header `X-API-Key`	Static key header, 3x retry policy
N7-04B	`Code_CheckTokenExpiry`	Code	Inspect static data cache against `Date.now()`	In-memory token expiration evaluator
N7-05	`IF_TokenIsExpired`	IF	Mode: Basic, Condition: `needsRefresh == true`	`{{ $json.needsRefresh }} == true`
N7-06	`HTTP_RefreshAuthToken`	HTTP Request	POST `/mock/auth/token`	Client credentials token exchange payload
N7-07	`Code_SaveFreshToken`	Code	Update `staticData.bearerToken` and `expiresAt`	Cache refresh mechanism
N7-08	`HTTP_CallWithBearerToken`	HTTP Request	GET protected resource with `Authorization: Bearer`	Dynamic Bearer header injection
N7-09	`Code_AuditAuthSuccess`	Code	Format execution verification output	Diagnostic audit summary

6. Verification Checkpoints & Testing Guide

Test 1: API Key Header Verification

1. Execute the workflow with `testMode: 0`.
2. `Switch_AuthPattern` routes directly to `HTTP_CallWithApiKey`.
3. Verify the remote server receives header `X-API-Key: key_live_prod_2026`.
4. Confirm response: HTTP 200 with status `"AUTHORIZED"`.

Test 2: Bearer Token Auto-Refresh Lifecycle (Cache Miss vs Cache Hit)

1. **Pass 1 — Cold Cache Execution (Cache Miss):**
   - Workflow static data is empty.
   - `Code_CheckTokenExpiry` evaluates `needsRefresh: true`.
   - Flow branches to `HTTP_RefreshAuthToken` and retrieves a fresh token (`jwt_mock_access_token_sec99182`).

- `Code_SaveFreshToken` stores the token and records expiration 3600 seconds forward.
- `HTTP_CallWithBearerToken` executes successfully returning HTTP 200.

2. **Pass 2 — Warm Cache Execution (Cache Hit):**
   - Execute the workflow a second time within 60 seconds.
   - `Code_CheckTokenExpiry` observes token is present and valid (`expiresAt - now > 60000`), evaluating `needsRefresh: false` .
   - Flow **bypasses the refresh node completely**, routing directly to `HTTP_CallWithBearerToken` using `$json.cachedToken` .
   - Result: Zero redundant token requests, eliminating API overhead.

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7. Production Risks & Mitigation Strategies

Production Risk	Operational Impact	Architectural Mitigation
Token Invalidation Before Expiry	Identity provider revokes token server-side, causing HTTP 401 despite client clock indicating validity.	Add reactive error trapping: on HTTP 401, invalidate cached token (`staticData.bearerToken = null`) and retry once.
Clock Drift Between Servers	Local server clock out of sync causes client to use token past actual expiration.	Enforce a conservative **60-second expiration buffer** (`expiresAt - now < 60000`).
Plain-Text Key Leakage	API keys or client secrets exposed in exported workflow JSON files.	Never hardcode raw production keys in node parameters; link to an encrypted n8n Credential object.
Network Flakiness on Key Auth	Temporary network timeout fails an API-key-authenticated request.	Configure `Retry on Fail` with exponential backoff (3 attempts, 2000 ms backoff) on HTTP Request nodes.

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8. LMS Submission Notes & Assessor Text

Assessor Summary

- **Topic:** Topic 7: API Integration Patterns & Authentication Methods
- **Repository Directory:** `https://github.com/3OOM/n8n-hands-on-level-2/tree/main/topic-07-api-integration-patterns-auth`
- **Workflow Source:** `Topic_7_API_Integration_Patterns_Auth.json`

Technical Challenges & Architectural Resolutions:

1. **Proactive In-Memory Token Caching:** Continuous token re-issuance exhausts OAuth2 rate limits and introduces latency. Resolved by persisting active tokens in n8n's workflow static memory (`$getWorkflowStaticData('global')`) paired with a 60-second proactive expiration buffer, executing token exchanges only when strictly required.
2. **Bypassing Redundant Refresh Roundtrips:** When cached tokens remain valid, the workflow bypasses `HTTP_RefreshAuthToken` via `IF_TokenIsExpired` and routes directly to the protected resource endpoint, cutting execution time in half.
3. **Resilient API Key Header Dispatch:** External API endpoints frequently experience transient network blips. Resolved by enabling `Retry on Fail` (3 retries with 2000 ms intervals) on `HTTP_CallWithApiKey` , ensuring

enterprise-grade reliability.